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MIL-P-16685D(NAVY)
30 April 1970
SUPERSEDING
MIL-P-16685C
12 October 1962
(See 6.4)

MILITARY SPECIFICATION

PACKING, MATERIAL AND PACKING PREFORMED

(STUFFING-TUBE FOR ELECTRIC CABLES)

All interested commands of the Department of the Navy have concurred in the use of this specification.

1. SCOPE

1.1 Scope. This specification covers rounded packing on spools or reels and preformed packing for electric cable seals in stuffing tubes.

1.2 Classification. Packing shall be of the following classes, compositions, and types, as specified (see table I, figures 1 and 2 and 6.2):

Class 1 - Rounded packing on spools or reels.
  Composition A - Retainer packing.
  Composition B - Sealing packing.
Class 2 - Preformed. (Made up from composition A and B packing (see 3.2.)
  Type I - For deck and bulkhead stuffing tubes (see Table III).
  Type IA - For deck and bulkhead stuffing tubes (see Table III).
  Type II - For equipment stuffing tubes (see Table III).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
  PPP-B-566 - Boxes, Folding Paperboard.
  PPP-B-636 - Box, Fiberboard.
  PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple Wall.
  PPP-B-676 - Boxes, Setup.
  PPP-C-96 - Cans, Metal, 28 Gage and Lighter.

MILITARY
  MIL-P-116 - Preservation, Methods Of.
  MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible.
  MIL-A-148 - Aluminum Foil.

STANDARDS

MILITARY
  MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
  MIL-STD-129 - Marking for Shipment and Storage.
  MIL-STD-147 - Palletized and Containerized Unit Loads, 40" x 48", 4-Way (Partial), Pallet Skids, Runners, or Pallet-Type Base.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 5330

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Freight Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

NATIONAL CLASSIFICATION BOARD
National Motor Freight Classification Rules

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., 1616 "P" Street, N.W., Washington, D.C. 20036.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Class 1, rounded.

3.1.1 Material.

3.1.1.1 Composition A. Packing shall be composed of asbestos rovings or yarns, either braid over braid or braided square, and treated with a waterproof compound.

3.1.1.2 Composition B. Packing shall be composed of an admixture of cork particles, graphite, lubricant and a plastic binder enclosed in a woven cotton lattice jacket.

3.1.2 Sizes. Packing shall be furnished in rounded cross section in sizes 1/8 to 1/2 inch, inclusive, in increments of 1/16 inch, as specified (see 6.2).

3.2 Class 2, preformed.

3.2.1 Material.

3.2.1.1 Piece 1. Piece 1 shall be made of packing conforming to class 1, composition A, as shown on figures 1, 2, and 3.

3.2.1.2 Piece 2. Piece 2 shall be made of packing conforming to class 1, composition B, which may be made in one continuous piece in the form of a tight helix, or may be cut into individual rings (see figures 1, 2, and 4). If made of individual rings, the cuts shall be displaced approximately 90 degrees from the cuts in the adjoining rings. The cut in each top and bottom retainer rings shall be similarly displaced from the cut in the adjoining sealing component.

3.2.2 Coating. After molding, packing shall receive a thin coat of a suitable quick-drying resin preparation. The coating shall be of such a nature that it will in no way affect the sealing properties of the class 1, composition B packing, when tested as specified in 4.4.

3.2.3 Sizes. Packing shall be of the sizes shown in table I and as specified (see 6.2).

Table I - Sizes.

Size	d 1/ Inch	Dimensions				
		outside diameter Inches	inside diameter Inches	h-I (type I) Inches	h-IA (type IA) Inches	h-II (type II) Inches
A.	3/16	0.750	0.375			
B.	3/16	.875	.480			
C.	3/16	1.000	.610			
D.	3/16	1.125	.719	1-3/8 ± 1/16	Same as h-I	1-1/8 ± 1/16
E.	3/16	1.125	.778			
F.	3/16	1.187	.812			
G.	1/4	1.375	.911			
J.	1/4	1.500	1.037			

See footnote at end of table.

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Table I - Sizes (cont'd).

Size	d 1/ Inch	Dimensions				
		outside diameter Inches	inside diameter Inches	h-I (type I) Inches	h-IA (type IA) Inches	h-II (type II) Inches
K.	1/4	1.625	1.120	2-5/8 ± 3/32	1-11/16 ± 1/16	1-3/8 ± 1/16
L.	1/4	1.687	1.232			
M.	1/4	1.875	1.375			
N.	1/4	1.937	1.474			
P.	1/4	2.062	1.586			
R.	5/16	2.187	1.681			
S.	3/8	2.562	1.820	4-5/8 ± 1/8	2-9/16 ± 3/32	2-1/8 ± 3/32
T.	3/8	2.750	2.030			
V.	3/8	2.875	2.142			
W.	3/8	3.000	2.280			
X.	3/8	3.125	2.427			
Y.	3/8	3.250	2.545			
Z.	3/8	3.437	2.673			

1/ Minimum diameter of size of piece 1 before molding.

3.3 Resistance to accelerated aging. Packing shall withstand the test specified in 4.4.1 without hardening or undergoing any change which indicates deterioration.

3.4 Molding. The rounded, class 1, packing shall be capable of being readily molded or prefabricated into packing sets. This capability shall be determined by the test described in 4.4.3.

3.5 Performance. Packing shall be capable of remaining tight indefinitely, and shall have no deleterious effects upon the sheathing of electric cables or upon the metallic armor of electric cables, where used against water, air, or gas at pressures from atmospheric to 25 pounds per square inch (p.s.i.g.) at temperatures up to 212°F. (see 4.4.2).

3.6 Compressibility. Packing shall be formed under the minimum pressures specified in table II with the packing and molds maintained at a minumum temperature of 100°F. When tested as specified in 4.4.3, the height shall be not less than 80 percent of the original height. For packing, class 1, furnished in bulk, spools or reels, check compressibility of all diameter of cable and packings in mold sizes equal to stuffing tube sizes C, L, R and V, respectively, in the above manner.

Table II - Molding pressures.

Type	Size	Minimum pressure (p.s.i. of annular area of packing)
I	A - J	600
I	K - Z	400
IA, II	A - R	600
IA, II	S - Z	400

3.7 Workmanship. Workmanship shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The inspection system which the supplier is required to maintain, as provided in the inspection clause of the contract or order, shall be in accordance with MIL-I-45208 (see 6.2).

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4.1.2 Test reporting. Test results shall be recorded by the supplier on test forms prepared by the supplier and approved by the Defense Contract Administration Service (DCAS). Provision shall be made for recording of all required data. Each test report form shall be identified by the number of this specification and by the particular test paragraph number and its title.

4.1.3 Test reports. All test data shall be approved by the DCAS prior to the shipment of the equipment. After approval, one file copy of all test data shall be forwarded to the agencies concerned. The data shall be bound into a 9 by 11-1/2 inch binder. The cover and title page shall give sufficient information to identify the equipment. A table of contents shall be included and list each test required by 4.3 and 4.4. No special test format will be required. However, each test shall be prefaced by the name of the test and applicable test paragraph. The data submitted shall be copies of the actual data taken on the test floor and not retyped data. The forms used shall allow sufficient columns to make instrument corrections and necessary calculations. The DCAS shall sign the cover sheet of the test report and note thereon the number of test sheets verified.

4.2 Sampling for quality conformance inspection.

4.2.1 Inspection lot. All packing of the same class, composition, type, or size shall be considered a lot for purposes of acceptance inspection.

4.2.2 Sampling for examination. Sample sets or spools shall be selected at random in accordance with MIL-STD-105 at Inspection Level I for the examination of 4.3.1. The Acceptable Quality Level shall be 2.5 percent defective.

4.2.3 Sampling for tests. Sample sets or spools shall be selected at random in accordance with MIL-STD-105 at Inspection Level S2 for tests specified in 4.3.2.

4.2.4 Sampling for compression test. A random sample of sets or spools shall be selected from each lot in accordance with 4.2.3 and subjected to the test specified in 4.4.3.

4.2.5 Sampling for production check test. Three yards of rounded packing and two sets of preformed packing from the first lot of each type and class and 1-1/2 yards of rounded packing and one set of preformed packing from every tenth lot shall be selected from the same class and type for the tests specified in 4.4. The samples shall be chosen so that over a period of time a representative selection of sizes will be sampled.

4.3 Quality conformance inspection.

4.3.1 Visual and dimensional examination. Each of the sample sets or spools (see 4.2.2), shall be visually and dimensionally examined to verify compliance with this specification. Any set or spool in the sample containing one or more visual or dimensional defects shall not be offered for delivery and if the number of defective sets in any sample exceeds the acceptance number for that sample, this shall be cause for rejection of the lot represented by the sample.

4.3.2 Quality conformance tests. Unless otherwise specified (see 6.2), quality conformance tests shall include the tests specified in 4.4 for each size, type and class of packing.

4.3.2.1 Compression test. Each of the sample sets or spools selected in accordance with 4.2.4 shall be tested in accordance with 4.4.3 to verify compliance with this specification. If any of the packing does not meet the requirements of this test, this shall be cause for rejection of the lot represented by the sample.

4.3.2.2 Production check test. The sample selected as specified in 4.2.5 shall be subjected to the tests specified in 4.4.

4.3.2.2.1 Action in case of failure. Lot acceptance and rejection will normally be on the basis of the sampling, examination and tests specified in 4.2, 4.3.1, and 4.3.2.1 only, and acceptance will not be withheld pending receipt of the test report on the production check test samples. However, upon receipt of an unsatisfactory report on any sample chosen for production check tests, additional samples shall be selected from every subsequent lot offered for delivery. These samples shall be subjected to the tests wherein failure was observed. Lots shall then be accepted only upon receipt of a satisfactory test report on the samples so selected. This additional testing shall be discontinued and lot acceptance returned to the normal basis when two successive lots have been accepted.

4.4 Test procedures.

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4.4.1 Accelerated aging test. Specimens shall be suspended in an electrically heated oven and shall be subjected to dry circulating air at 200°F. for 1000 hours without hardening or softening or without changing in any other respect which will indicate that the material is deteriorating.

4.4.2 Performance test. This test shall be conducted in representative sizes of bulkhead stuffing tubes. Both plain and armored cable (including sheathing material of polychloroprene, impervious resins and thermoplastic resins) shall be used during the test. After the packing has been installed in the tubes, the gland units shall be drawn up snugly with a wrench 20 inches in length. An air pressure of 25 p.s.i.g. for bulkhead stuffing tubes shall then be applied within the apparatus and any leakage that occurs from the tubes shall be noted and recorded. The air, with pressure still applied within the apparatus, shall be heated to a temperature of 212°F. by means of a water bath surrounding the compartments. These conditions shall be maintained for 1-1/2 hours and any leak which occurs from the tubes shall be noted and recorded. At the end of this time the apparatus shall be allowed to cool and, when cold, the leakage from the tubes shall again be recorded. The above conditions shall be repeated at least three times. On conclusion of the test, the packing shall be removed from the tubes and the cables examined for crimpage, deformation, or evidence of gross damage by chemical attack of armor or cable sheath. If any of these mentioned defects are in evidence, this shall be cause for rejection of the lot represented by the sample.

4.4.3 Compression test. The adequacy of the molding pressures, as specified in 3.6 shall be determined as follows: Measure the height dimension (h-I, h-IA or h-II of figures 1 and 2, as applicable) of a sample of preformed packing. Place the sample in the corresponding size mold and subject it to the pressure specified in 3.6 for a period of 3 minutes. Release pressure and measure the same dimension as soon as the set is removed from the mold.

4.5 Inspection of preparation for delivery. Preservation-packaging, packing, and marking shall be inspected to determine conformance with Section 5 of this specification.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in Section 2, see 6.3).

5.1 Preservation-packaging. Preservation-packaging shall be Level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Packing on spools and reels.

5.1.1.1.1 Preservation-packaging. Each spool or reel shall be packaged Method IC-1 of MIL-P-116 with a greaseproof barrier material conforming to MIL-B-121 or MIL-A-148 at the suppliers option. The greaseproof barrier material specified herein is not required when the unit container selected (see 5.1.1.1.1.1) is treated for greaseproofness or is inherently greaseproof.

5.1.1.1.1.1 Unit containers. Each spool or reel, packaged in accordance with 5.1.1.1.1 shall be placed in a paperboard or fiberboard box or a metal can at the suppliers option. The unit container selected shall conform to the applicable following specification.

Specification	Container
PPP-B-566	Box, paperboard, folding
PPP-B-636	Box, fiberboard
PPP-B-676	Box, paperboard, set up
PPP-C-96	Can, metal, 28 gage and lighter

Sealing and closure, as applicable, of the unit container shall conform to the applicable container specification or appendix thereto.

5.1.1.2 Preformed. Preformed packing shall be preserved-packaged Level A in accordance with MIL-P-4861.

5.1.2 Level C. Preservation-packaging shall be sufficient to afford protection against damage during shipment from the supply source to the first receiving activity for immediate use.

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5.2 Packing. Packing shall be Level A, B, or C as specified (see 6.2).

5.2.1 Level A. Packing, preserved-packaged as specified, shall be packed in containers conforming to any one of the following specifications at the contractors option:

Specification	Container	Type or class
PPP-B-636	Fiberboard	Weather-resistant
PPP-B-640	Fiberboard, corrugated, triple-wall	Class 2

Closure shall be in accordance with the appendix of the applicable box specification except that adhesives shall not be used. Containers shall be reinforced with tape banding in accordance with the appendix to the applicable container specification. Containers shall not exceed the weight limitations of the applicable container specification. Unit containers conforming to PPP-B-636 meeting the requirements for this level of packing may be shipped without further packing provided there is only one container being shipped to a destination and the requirements for reinforcement are met.

5.2.2 Level B. Packing, preserved-packaged as specified, shall be packed in containers conforming to any of the following specifications at the contractors option:

Specification	Container	Type or class
PPP-B-636	Fiberboard	Domestic
PPP-B-640	Fiberboard, corrugated triple-wall	Class 1

Closure shall be in accordance with the appendix of the applicable box specification except that adhesives shall not be used. Containers shall not exceed the gross weight limitations of the applicable container specification. Unit containers conforming to PPP-B-636 meeting the requirements for this level of packing may be shipped without further packing provided there is only one container being shipped to a destination and the requirements for reinforcement are met.

5.2.3 Level C. Packing, preserved-packaged as specified, shall be packed in containers in a manner acceptable to the carrier at the lowest rate, which will afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers or method of packing shall conform to the Uniform Freight Classification, National Motor Freight Classification or other carrier Rules and Regulations as applicable to the mode of transportation. When the unit container meets the applicable carrier rules it may be used as the shipping container.

5.3 Palletized and containerized unit loads. When specified (see 6.2) packing, preserved-packaged and packed as specified shall be palletized or containerized in accordance with MIL-STD-147.

5.4 Marking. In addition to any special marking required (see 6.2), interior packages, exterior shipping containers and palletized or containerized unit loads shall be marked for shipment in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The packing covered by this specification is intended for use for stuffing tubes for electric cables (see table III).

Table III - Application of packing types, class 2.

Stuffing tubes		Plugs and outlets	
NAVSHIPS drawing or specification number	Packing type required	NAVSHIPS drawing or specification number	Packing type required
MIL-S-24235/9 MIL-S-24235/10 MIL-S-24235/12	I	9-S-4799-L 9000-S6202-74438 MIL-S-24235/11	II size B
9-S-4436-L, ALT 9 or later	I	MIL-S-24235/13	II
9-S-5166-L ALT 10 or later	I	MIL-S-24235/14	II
9-S-5166-L ALT 9 or earlier	IA		

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6.1.1 Class 1. All references to Class 1, part A or B which were listed in MIL-P-16685C have been replaced by Class 1, composition A or B in MIL-P-16685D(NAVY).

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Class, composition, type and size required (see 1.2, 3.1.2, and 3.2.3).
(c) Inspection responsibilities and facilities (see 4.1).
(d) If a manufacturer's respective size, type, and class of packing have passed all tests on a previous submission and the materials, dimensions and method of manufacture have not changed, none of these tests under section 4 shall be repeated, except at the discretion of the Government inspector; the manufacturer shall submit a copy of the test report as evidence that his previous deliveries of packing material of the specific size, type, and class under order have passed the tests of section 4 (see 4.3.2).
(e) Level of preservation-packaging and packing required (see 5.1 and 5.2).
(f) When palletization and containerization are required (see 5.3).
(g) Special marking, if required (see 5.4).

6.3 Sub-contracted material. The preparation for delivery requirements of referenced documents listed in Section 2 do not apply when material is procured by the supplier for incorporation into the equipment and which lose their separate identity when the equipment is shipped.

6.4 CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

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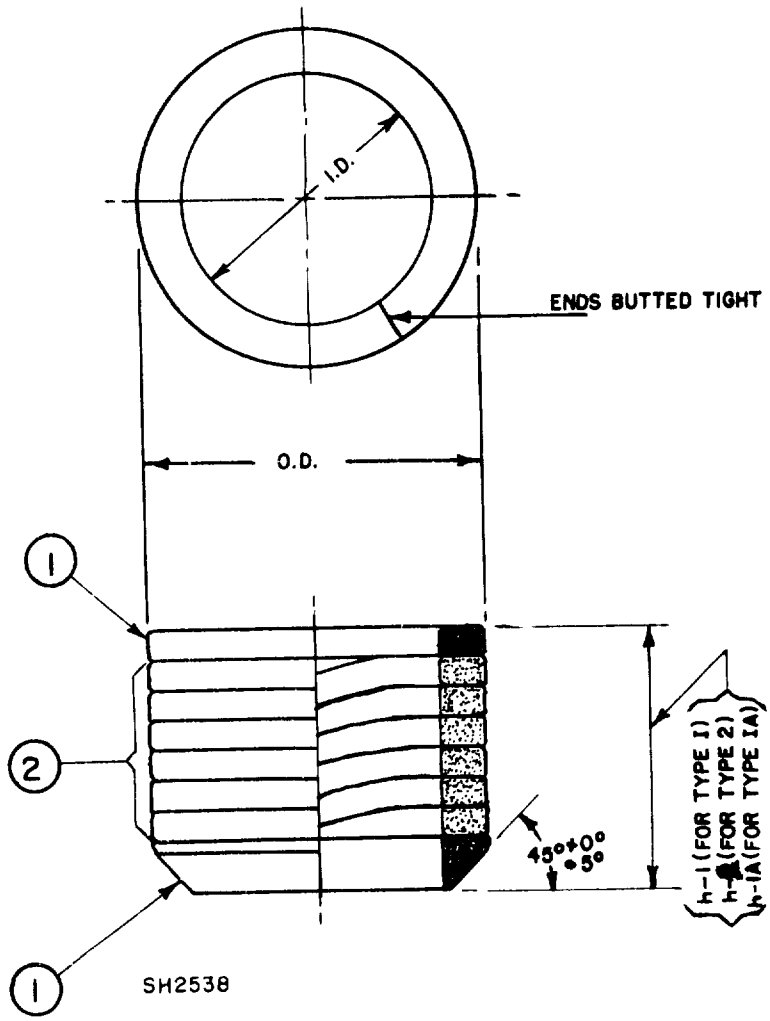


Figure 1 - Type I sizes (A to J).
Type IA (all sizes).
Type II (all sizes).

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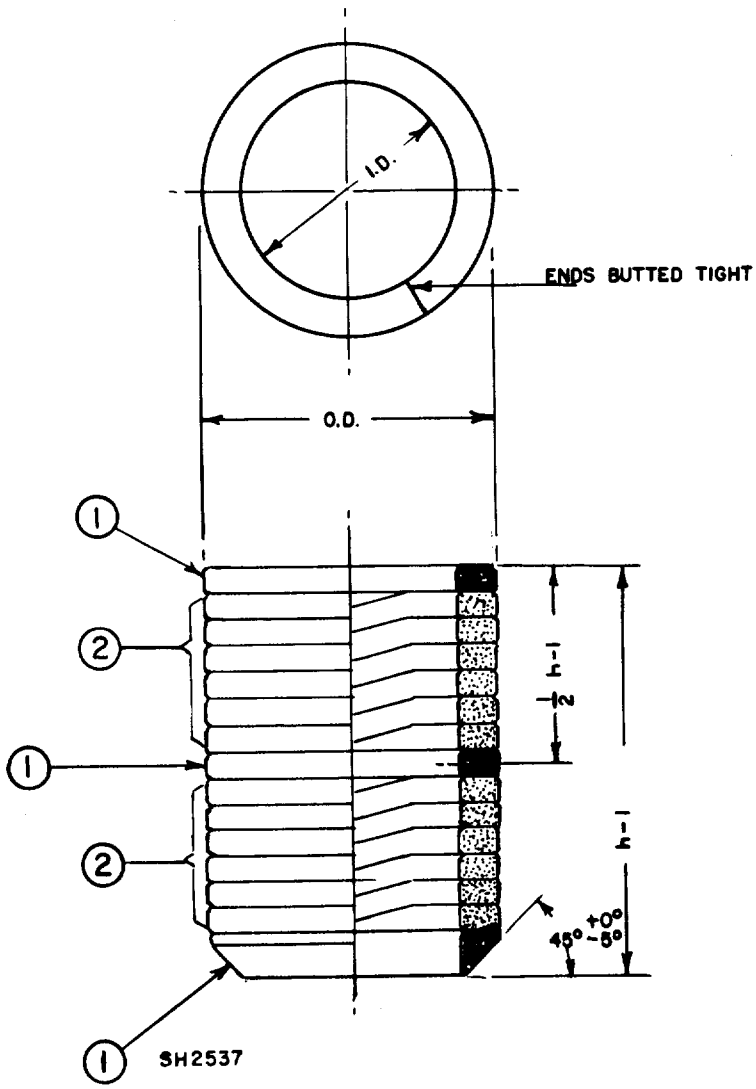


Figure 2 - Type I (sizes K to Z).

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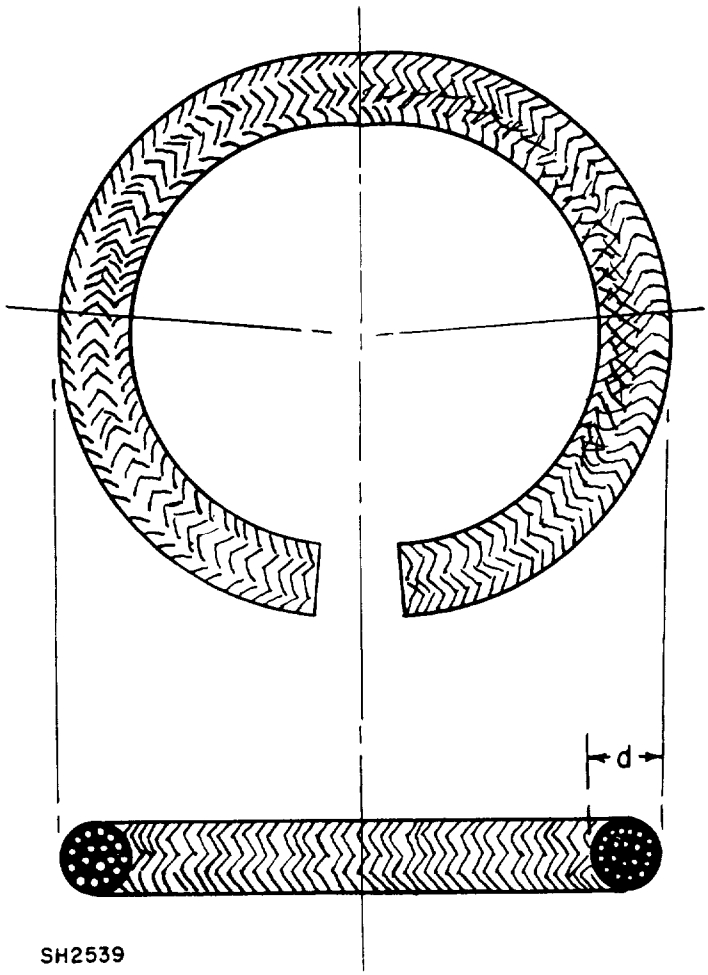
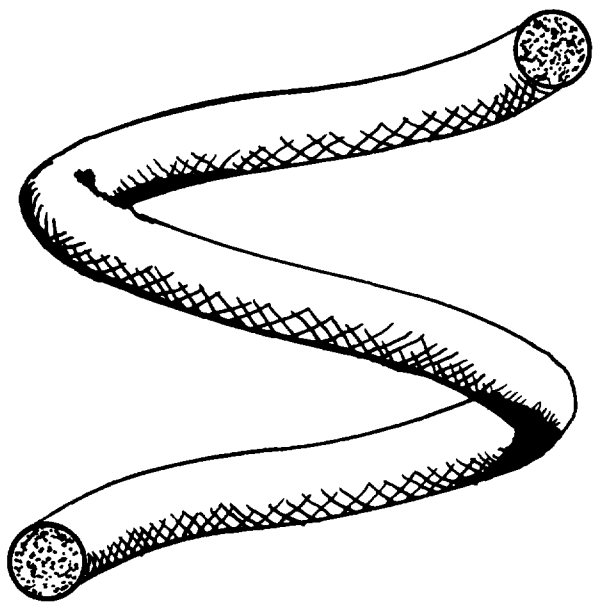


Figure 3 - Piece 1.

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Figure 4 - Piece 2.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD, FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE 0101-807-2000

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